

Homework – SMPC Assignment 1

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April 21, 2026

1a: Ideal Voltage Regulator

- **Given Parameters:** Input $V_{in} = 1 \text{ V}$, $V_{bat} = 5 \text{ V}$, $R_{out} = 4 \Omega$
Target $V_{out} = 1.8 \text{ V}$
- **Calculated ideal efficiency η :**

$$\eta = \frac{V_{out}}{V_{bat}} = \frac{1.8}{5} \cdot 100 = 36 \%$$

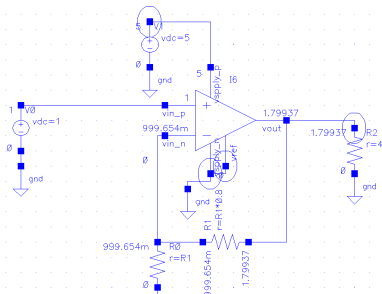
1a: Voltage Regulator

- The voltage regulator is realized in Cadence by the ideal non-inverting OPAMP circuit shown in the Figure below.

- **Simulated efficiency η :**

$$\eta = \frac{V_{out} \cdot I_{out}}{V_{bat} \cdot I_{bat}} = 35.98 \%$$

- The voltage regulator provides stable and linear voltage, but it comes at the cost of low efficiency if $V_{out} \ll V_{bat}$. Therefore, this topology is not a good choice for this application.



2a: Duty Cycle and Output Voltage

- **Given Parameters:** Input $V_{in} = 5V$, Target $V_{out} = 1.8V$, $f_s = 2MHz$ ($T = 500\text{ ns}$).
- **Calculated Duty Cycle (D_{ideal}):**

$$D_{ideal} = \frac{V_{out}}{V_{in}} = \frac{1.8V}{5V} = 0.36$$

- **Theoretical Output Voltage:**

$$V_{out} = D \cdot V_{in} = 0.36 \cdot 5V = 1.8V$$

(Note: In simulation, D was adjusted to 0.382 to compensate for conduction losses.)

- **Simulated V_{out} :** 1.805V

2a: Simulated V_{out}

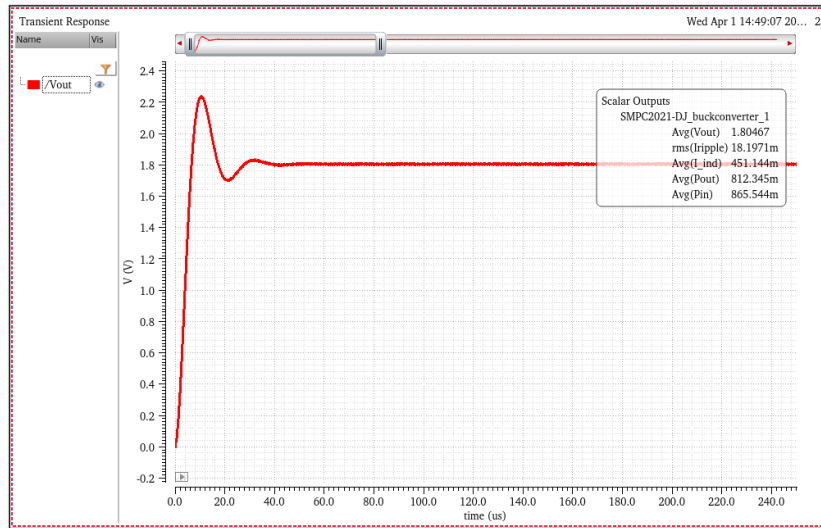


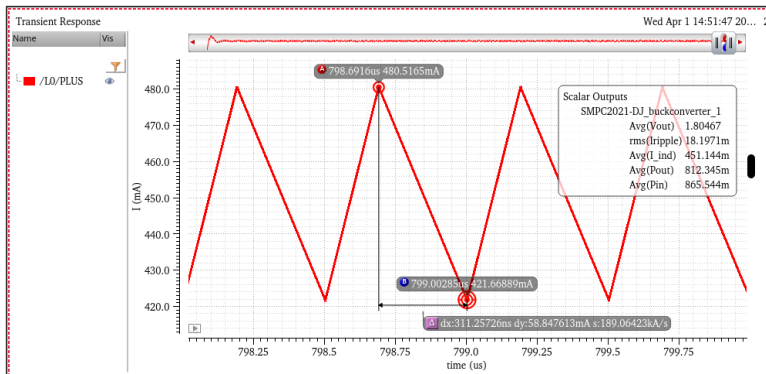
Figure: Simulated output voltage with $D = 0.382$

2b: Inductor Peak-to-Peak Ripple Current

- **Calculated Ripple Current (ΔI_L):** Using $V_{in} = 5V$, $V_{out} = 1.8V$, $L = 10\mu H$, and $T_{on} = 0.369 \cdot 500ns = 184.5ns$:

$$\Delta I_L = \frac{V_{in} - V_{out}}{L} \cdot T_{on} = \frac{5 - 1.8}{10 \cdot 10^{-6}} \cdot 184.5 \cdot 10^{-9} = 59.04mA$$

- **Simulated Ripple Current: 58.8 mA**



2c: Conduction Loss

- **Necessary Parameters (from 2.0 MHz simulation):**

- PMOS R_{on} : 165 m Ω
- NMOS R_{on} : 180 m Ω
- Average Inductor Current: $I_{L,avg} = 451$ mA
- RMS Inductor ripple current: $I_{L,ripple,rms} = 18.2$ mA
- Duty Cycle (D): 0.382

- **Calculated Conduction Loss (P_{cond}):**

- Effective Resistance: $R_{eff} = R_{on,p}D + R_{on,n}(1 - D) = 174.27$ m Ω
- $P_{avg} = I_{L,avg}^2 \cdot R_{eff} = 35.45$ mW
- $P_{ripple} = I_{L,ripple,rms}^2 \cdot R_{eff} = 0.06$ mW
- **Total Conduction Loss:**
- **$P_{cond} = 35.51$ mW**

2c: Power Transistor Sizing

- **Design Goal:** Target a conduction loss of $\approx 5\%$ of total power.
- **Current Status:** $P_{cond} \approx 4.37\%$ (Under-utilized loss budget).
- **Proposed Strategy: Downsizing Transistor Width (W)**
 - Conduction loss is dominated by R_{on} , where $R_{on} \propto \frac{L}{W}$.
 - To increase P_{cond} to exactly 5%, the transistor widths W_{pmos} and W_{nmos} should be reduced by a factor of $\approx \frac{4.37}{5.0} = 0.87$.
- **Motivation for Downsizing:**
 - **Area Efficiency:** Smaller transistors reduce the total silicon area, lowering manufacturing costs.
 - **Switching Loss Improvement:** Reducing W decreases the gate capacitance (C_{gg}).
 - This leads to lower **switching losses** ($P_{sw} = C_{gg} V^2 f_{sw}$), increasing the overall converter efficiency despite the intentionally higher conduction loss.
- **Conclusion:** By shrinking the transistors by a factor of ≈ 0.87 to meet the 5% conduction limit, we optimize the balance between static (R_{on}) and dynamic (C_{gg}) losses.

2c: Conduction Loss (Post-Resizing @ 2.0 MHz)

- **Updated Parameters (from resized simulation):**

- PMOS R_{on} : 191 m Ω
- NMOS R_{on} : 207 m Ω
- Average Inductor Current: $I_{L,avg} = 447.84$ mA
- RMS Inductor ripple current: $I_{L,ripple,rms} = 17.99$ mA
- Average Output Power: $P_{out} = 800.46$ mW
- Duty Cycle (D): 0.382

- **Calculated Conduction Loss (P_{cond}):**

- Effective Resistance: $R_{eff} = R_{on,p}D + R_{on,n}(1 - D) = 200.89$ m Ω
- $P_{avg} = I_{L,avg}^2 \cdot R_{eff} = 40.29$ mW
- $P_{ripple} = I_{L,ripple,rms}^2 \cdot R_{eff} = 0.07$ mW
- **Total Conduction Loss:**
- **$P_{cond} = 40.36$ mW ($\approx 5.04\%$ of P_{out})**

- **Full Load Efficiency:**

- Total Input Power: $P_{in} = 853.46$ mW
- **Power Efficiency (η): 93.79%**

2d: Gate Driver Sizing (Calculations)

- **Design Objective:** Achieve a transition time (t_{trans}) of ≈ 2.0 ns.
- **Extracted Gate Capacitances:**
 - PMOS C_{gg} : 23.28 pF
 - NMOS C_{gg} : 15.03 pF
- **Required Driver Current (I_{drv}):**
 - Using $I = C \frac{\Delta V}{\Delta t}$, assuming a 5V gate swing:
 - $I_{drv,pmos} = 23.28 \text{ pF} \cdot \frac{5\text{V}}{2.0\text{ns}} = \mathbf{58.20 \text{ mA}}$
 - $I_{drv,nmos} = 15.03 \text{ pF} \cdot \frac{5\text{V}}{2.0\text{ns}} = \mathbf{37.58 \text{ mA}}$
- **Sizing Strategy:**
 - The initial unoptimized transition time was measured at **250** ps.
 - To reach the target of 2.0 ns, the driver strength must be reduced.
 - While a linear scaling of $0.125\times$ was theoretically calculated, simulation shows a non-linear response where the driver becomes too weak (15.4 ns).
 - **Adjustment:** The inverter widths are being iteratively tuned ($0.6\times$) to achieve the optimal 2.0 ns window.

2d: Driver Motivation & Shoot-Through Mitigation

- **Why a 2.0 ns Transition Time?**

- **Loss Trade-off:** Faster transitions minimize the $V - I$ overlap period, reducing switching losses (P_{sw}).
- **Practical Limit:** Pushing significantly faster than 2.0 ns requires excessively large gate drivers. This increases the "driving-the-driver" losses and consumes unnecessary silicon area.
- 2.0 ns represents an optimal balance for a 2.0 MHz switching frequency.

- **Theoretical Mitigation of Shoot-Through:**

- **The Risk:** In a CMOS bridge, if the PMOS turns ON before the NMOS is fully OFF, a direct V_{dd} -to-GND short-circuit (shoot-through) occurs.
- **Consequences:** This causes massive current spikes, EMI issues, and severely degrades efficiency.
- *Note: The current design successfully avoids shoot-through by utilizing appropriate delay offsets in the gate signals.*

2d: Gate Driver Optimization Results

- Simulated Waveforms (Post-Sizing & Mitigation):

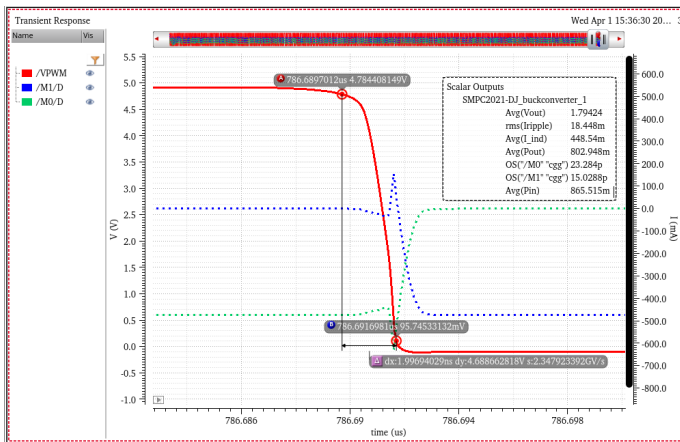


Figure: Transition time of 2.0 ns with no shoot through

2e & 2f: 16Ω Load Analysis (Light Load)

- **Updated Parameters (16Ω Load @ 2.0 MHz):**

- PMOS R_{on} : $188\text{ m}\Omega$ — NMOS R_{on} : $204\text{ m}\Omega$
- Average Inductor Current: $I_{L,avg} = 112.56\text{ mA}$
- RMS Inductor ripple current: $I_{L,ripple,rms} = 18.04\text{ mA}$
- Average Output Power: $P_{out} = 203.98\text{ mW}$
- Duty Cycle (D): 0.370

- **Calculated Conduction Loss (P_{cond}):**

- Effective Resistance: $R_{eff} = R_{on,p}D + R_{on,n}(1 - D) = 198.08\text{ m}\Omega$
- $P_{avg} = I_{L,avg}^2 \cdot R_{eff} = 2.51\text{ mW}$
- $P_{ripple} = I_{L,ripple,rms}^2 \cdot R_{eff} = 0.06\text{ mW}$
- **Total Conduction Loss: $P_{cond} = 2.57\text{ mW}$ ($\approx 1.26\%$ of P_{out})**

- **Efficiency & Dominant Loss:**

- **Power Efficiency (η): 93.89% ($P_{in} = 217.24\text{ mW}$)**
- **Dominant Loss: Switching Loss.**
- At this light load, conduction loss scales with I^2 and becomes negligible (2.57 mW).
- The frequency-dependent switching losses (gate drive and $V - I$ overlap) now account for the majority ($\approx 10.69\text{ mW}$) of the total 13.26 mW loss.